

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) **Marks****Q. 1 Solve Any Two of the following.****12**

- A) Explain in brief FSM Properties and Limitations? Write Applications of FSM?

Understand / **6**
CO-1

- B) Construct Mealy Machine for the given Moore Machine?

Apply/ CO-2 **6**

Present state	Next State		Output
	a=0	a=1	
→q1	q1	q2	0
q2	q1	q3	0
q3	q1	q3	1

- C) Design a FSM to check whether a given decimal number is divisible by 5?

Understand/ **6**
CO-1**Q.2 Solve Any Two of the following.****12**

- A) Explain in brief Chomsky hierarchy with suitable examples?

Understand / **6**
CO-2

- B) Eliminate NULL production from the grammar G given below:

Apply / CO-3 **6**

a. $S \rightarrow a / Xb / aYa$

b. $X \rightarrow Y / \wedge$

c. $Y \rightarrow b / X$

Write the productions after elimination.

- C) Check whether the given grammar is ambiguous or not-

Analyse / **6**
CO-3

a. $S \rightarrow a / abSb / aAb$

b. $A \rightarrow bS / aAAb$

Q. 3	Solve Any Two of the following.		12
A)	Consider a left linear grammar as given below – a. $S \rightarrow Sa Abc$ b. $A \rightarrow Sa Ab a$ Convert left linear grammar to right linear grammar.	Apply / CO-3	6
B)	Convert the given to context free grammar to CNF forms a. $S \rightarrow ASB$ b. $A \rightarrow aAS a \epsilon$ c. $B \rightarrow SbS A bb$	Apply / CO-3	6
C)	Derive a derivation tree for the string aabbabba for the given context free grammar (CFG) – a. $S \rightarrow aB bA$ b. $A \rightarrow a aS bAA$ c. $B \rightarrow b bS aBB$	Analyse / CO-3	6
Q.4	Solve Any Two of the following.		12
A)	Define a PDA & list three important properties of a PDA Machine?	Understand / CO-2	6
B)	Construct a PDA to check well formedness of parenthesis?	Create / CO-3	6
C)	Convert the following grammar to a PDA that accepts the same language. a. $S \rightarrow 0S1 A$ b. $A \rightarrow 1A0 S \epsilon$	Apply / CO-3	6
Q. 5	Solve Any Two of the following.		12
A)	Design a TM for an equal number of a's and b's must follow a.	Create / CO-3	6
B)	Design a TM which recognizes palindromes over $\Sigma = \{a, b\}$	Create / CO-3	6
C)	Explain 'Halting Problem of turning machine' with neat diagrams?	Understand / CO-4	6

*** End ***